“Pandas CSV Reader and Basic Analysis of Impact of Advertising on Sales”

A PROJECT SUBMITTED TO

SYNTECXHUB Technologies

Kanpur Nagar, Uttar Pradesh

FOR THE INTERNSHIP

IN

DATA SCIENCE

By

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07-04-2026

Introduction

In the modern digital era, data has become one of the most valuable assets for organizations across all industries. Businesses increasingly rely on data analysis to make informed decisions, optimize their strategies and improve overall performance. In particular, the field of marketing and advertising heavily depends on data to evaluate the effectiveness of different promotional channels and to maximize return on investment.

Advertising plays a crucial role in influencing customer behavior and driving sales. Companies invest significant amounts of money in various advertising platforms such as television, radio and newspapers. However, understanding which medium contributes the most to sales is essential for efficient budget allocation and strategic planning. This is where data analysis becomes highly useful.

This project focuses on analyzing an advertising dataset using Python and the Pandas library. The dataset contains information about advertising expenditures on TV, Radio and Newspaper, along with the corresponding sales figures. By examining this data, we aim to understand the relationship between advertising spend and sales performance.

The analysis involves loading and exploring the dataset, inspecting its structure and performing statistical computations to summarize the data. Techniques such as filtering, slicing and sub setting are applied to extract relevant information and focus on specific aspects of the dataset. These operations help in identifying patterns, trends and relationships among the variables.

Furthermore, the project demonstrates how basic data analysis techniques can be effectively used to derive meaningful insights from raw data. It also provides hands on experience in using Pandas for data manipulation and analysis, which is a fundamental skill in the field of data science.

Overall, this project serves as an introduction to real-world data analysis, highlighting the importance of data-driven decision making in advertising and business contexts. It lays a strong foundation for more advanced analytical techniques and future data science projects.

Objectives

- To read and import data from a CSV file using the Pandas library in Python.
- To inspect and understand the dataset using functions such as head(), tail() and info().
- To analyze the dataset by calculating summary statistics including mean, minimum, maximum and count.
- To filter and select relevant data from the dataset in order to extract meaningful information.
- To apply slicing and subsetting techniques for focused and detailed data analysis.

Methodology

The methodology of this project follows a structured approach to analyze the advertising dataset using the Pandas library in Python.

Data Source:

The dataset used in this project is an advertising dataset containing information about marketing expenditures on different platforms such as TV, Radio and Newspaper, along with the corresponding sales. This dataset is commonly used for educational and analytical purposes to understand the impact of advertising on sales performance.

Variables Description:

The dataset consists of 200 records and 4 columns, which include : TV, Radio, Newspaper and Sales. These variables represent the amount spent on different advertising media and the resulting sales generated, providing a clear understanding of the relationship between advertising budget and sales outcomes.

Process:

First, the dataset was loaded from a CSV file into a Pandas DataFrame for structured analysis. The dataset was then inspected using functions such as head(), tail(), dtypes, and info() to understand its structure and verify the data types.

After inspection, statistical analysis was performed using functions like mean(), median(), min(), max(), and count() to summarize the data and observe patterns in advertising spend and sales.

Further, data filtering techniques were applied to extract records based on specific conditions, such as higher TV advertising expenditure. Column selection was used to focus on relevant variables, and slicing techniques were applied to analyze specific subsets of the dataset.

Finally, the filtered and processed data was saved into a new CSV file for further use. This step-by-step methodology ensured efficient data handling and helped in extracting meaningful insights from the dataset.

Data Analysis

- **To read and import data from a CSV file using Pandas**

Code:

```
import pandas as pd

df = pd.read_csv("C:\\Users\\Dell\\Documents\\advertising.csv")

print(df.head())
```

Output:

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	12.0
3	151.5	41.3	58.5	16.5
4	180.8	10.8	58.4	17.9

- **To inspect the dataset using functions like head(), tail() and info()**

Code:

```
# Display first 5 rows

print("First 5 rows:")

print(df.head())


# Last 5 rows

print("\nLast 5 rows:")

print(df.tail())


# Data types

print("\nData Types:")

print(df.dtypes)


# Dataset info

print("\nDataset Info:")

print(df.info())
```

Output:

First 5 rows:

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	12.0
3	151.5	41.3	58.5	16.5
4	180.8	10.8	58.4	17.9

Last 5 rows:

	TV	Radio	Newspaper	Sales
195	38.2	3.7	13.8	7.6
196	94.2	4.9	8.1	14.0
197	177.0	9.3	6.4	14.8
198	283.6	42.0	66.2	25.5
199	232.1	8.6	8.7	18.4

Data Types:

TV	float64
Radio	float64
Newspaper	float64
Sales	float64
dtype: object	

Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 200 entries, 0 to 199

Data columns (total 4 columns):

	Column	Non-Null Count	Dtype
0	TV	200 non-null	float64
1	Radio	200 non-null	float64
2	Newspaper	200 non-null	float64
3	Sales	200 non-null	float64

dtypes: float64(4)

memory usage: 6.4 KB

- **To calculate summary statistics such as mean, minimum and maximum**

Code:

```
# Summary statistics
print("\nSummary Statistics:")
print(df.describe())

# Mean
print(df.mean(numeric_only=True))

# Minimum values
print(df.min(numeric_only=True))

# Maximum values
print(df.max(numeric_only=True))

# Count
print(df.count())
```

Output:

Summary Statistics:

	TV	Radio	Newspaper	Sales
Count	200.00	200.00	200.0	200.00
Mean	147.04	23.26	30.55	15.13
Std	85.85	14.84	21.7	5.28
Min	0.70	0.00	0.30	1.0
25%	74.37	9.97	12.75	11.00
50%	149.75	22.90	25.75	16.00
75%	218.82	36.5	45.1	19.05
max	296.40	49.60	114.00	27.00

- **To filter and select relevant data from the dataset**

Code:

```
# Load dataset
df = pd.read_csv("C:\\Users\\Dell\\Documents\\advertising.csv")

# Filter rows where TV advertising budget is greater than 200
filtered_data = df[df['TV'] > 200]

print("Filtered Data (TV > 200):")
print(filtered_data)

# Select only relevant columns
selected_data = filtered_data[['TV', 'Radio', 'Sales']]

print("\nSelected Relevant Columns:")
print(selected_data.head())
```

Output:

Filtered Data (TV > 200)

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
11	214.7	24.0	4.0	17.4
14	204.1	32.9	46.0	19.0
17	281.4	39.6	55.8	24.4
20	218.4	27.7	53.4	18.0

	TV	Radio	Newspaper	Sales
184	253.8	21.3	30.0	17.6
185	205.0	45.1	19.6	22.6
188	286.0	13.9	3.7	20.9
198	283.6	42.0	66.2	25.5
199	232.1	8.6	8.7	18.4

[67 rows x 4 columns]

Selected Relevant Columns:

	TV	Radio	Sales
0	230.1	37.8	22.1
11	214.7	24.0	17.4
14	204.1	32.9	19.0
17	281.4	39.6	24.4
20	218.4	27.7	18.0

- **To perform slicing and subsetting for focused analysis**

Code:

```
# First 10 rows
subset1 = df.iloc[0:10]
print("\nFirst 10 rows:")
print(subset1)
# First 5 rows and selected columns
subset2 = df.iloc[0:5][['Order ID', 'Product', 'Total Sales']]
print("\nSubset (First 5 rows with selected columns):")
print(subset2)
```

Output:

First 10 rows:

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	12.0
3	151.5	41.3	58.5	16.5
4	180.8	10.8	58.4	17.9
5	8.7	48.9	75.0	7.2
6	57.5	32.8	23.5	11.8
7	120.2	19.6	11.6	13.2
8	8.6	2.1	1.0	4.8
9	199.8	2.6	21.2	15.6

Subset (First 5 rows with selected columns):

	TV	Sales
0	230.1	22.1
1	44.5	10.4
2	17.2	12.0
3	151.5	16.5
4	180.8	17.9

Overall Conclusion

In this project, the advertising dataset was successfully analyzed using Python and the Pandas library. The dataset was explored to understand its structure, data types and overall composition. Various statistical measures such as mean minimum, maximum and count were calculated to summarize the data and identify key patterns.

Data filtering and selection techniques were applied to extract meaningful information, particularly focusing on records with higher advertising expenditure. Slicing and subsetting methods further helped in analyzing specific portions of the dataset in a more detailed manner.

The analysis indicates that advertising expenditure, especially on TV and Radio, has a noticeable impact on sales, suggesting a positive relationship between marketing efforts and revenue generation. The processed data was also successfully exported into a new CSV file for further use.

Overall, this project provided practical experience in data handling, analysis and manipulation using Pandas and demonstrated how basic data analysis techniques can be used to derive useful business insights.